

Transcriptomics in non-model organisms and ^{few} insights into metatranscriptomics

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Why meta/transcriptomics in a metagenomic workshop.1?

Genes: functional potential

Transcripts: **expressed genes**

This **adds value** to what we have predicted at the genomic level,
(but we are not studying the proteome)

However, given that transcription is energetically demanding,
the transcript detection is strong evidence of its use.

Why meta/transcriptomics in a metagenomic workshop.2?

Also, **functional redundancy** is often seen in microbial communities

Which organism(s) actively support these communities?

Mapping transcripts to metagenomic contigs (or even MAGs) can help to resolve the issue.

What are model organisms?

Species studied to understand biological processes that also apply to other organisms

- easy to work with
- reproduce quickly
- share important similarities with more complex organisms.

Some examples of model organisms

- Mouse
- Zebrafish
- *Drosophila melanogaster*
- *Arabidopsis thaliana*
- Yeast
- *E. coli*

And, what about non-model organisms?

Organisms studied for:

- basic research
(i.e., evolutionary studies)
- biodiversity and conservation
(i.e., Darwin Tree of Life Consortium)
- economic interests
(i.e., agriculture/biotechnology)

The extent of the research into non-model organisms varies

Usually,

the **genome** of the organism is obtained, together with

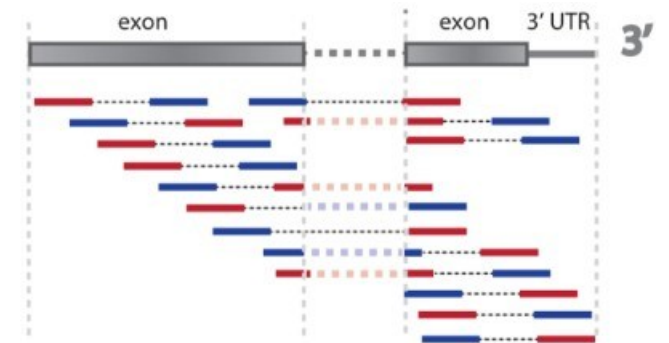
- gene prediction
- annotation

plus, in most cases, **RNASeq experiments** (short reads)



Refinement of gene models

Differential gene expression



What if the organism's genome is highly fragmented?
(typical with short reads)

Mapping rate of RNASeq data ↓

Possible solution

building the transcriptome from scratch, but which
sequencing platform should I use?

Hybrid approach (long reads plus short reads)

Long reads: *de novo* transcriptome (with isoforms)

Short reads: transcript sequence correction*
&
differential expression profiles

Our approach.1

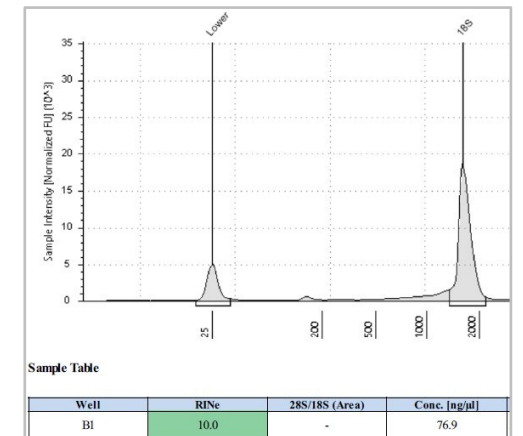
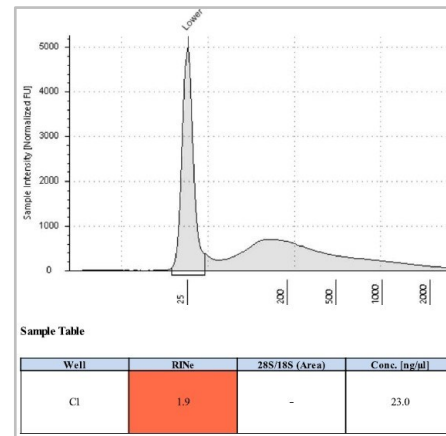
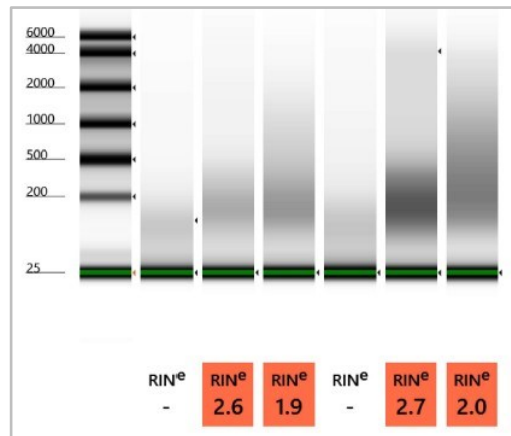
Acartia tonsa (copepod):

exposure to neonicotinoids

Lanice conchilega (polychaete):

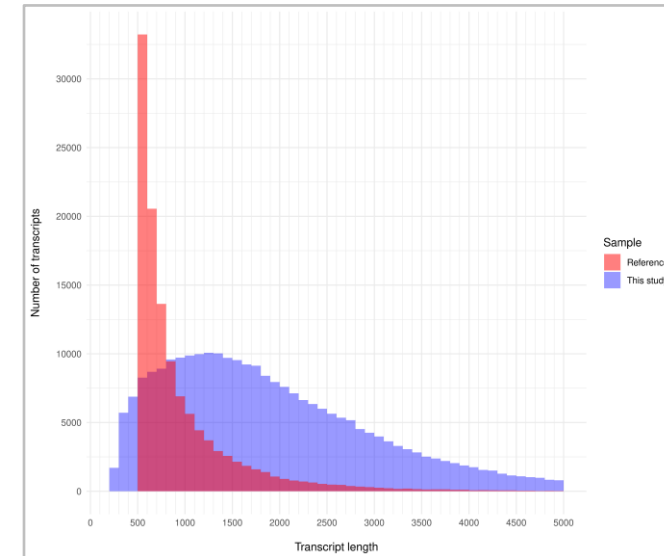
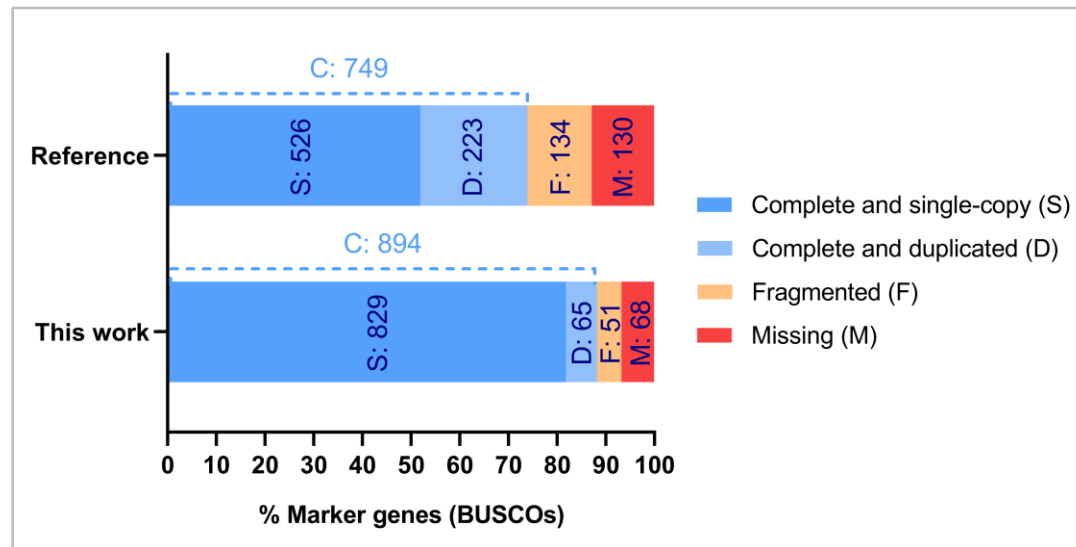
low frequency sound stress

- Sample collection and storage (RNA Protect first, then -80°)
- RNA extraction and quality check (RIN value)



Our approach.2

- Sample pool for sequencing with Oxford Nanopore
- RNASeq of individual samples using the Illumina platform (biological replicates for statistical analyses)
- *De novo* transcriptome with error correction and annotation



- RNASeq mapping and differential gene expression analysis (stay tuned...)

What about metatranscriptomics?

Hard task in the environment as it depends on:

- Community complexity
(few species? many species?)
- Type of sample under study
(water? sediment?)

(Size fractionation might be a solution for water samples,
but what about the global picture of the community?)

A possible approach for metatranscriptomics studies

- 1) Metagenomic data production
(MAGs & metagenomic contigs)
- 2) Gene prediction and annotation where possible
- 3) Metatranscriptomic data production and mapping
on the metagenomic data

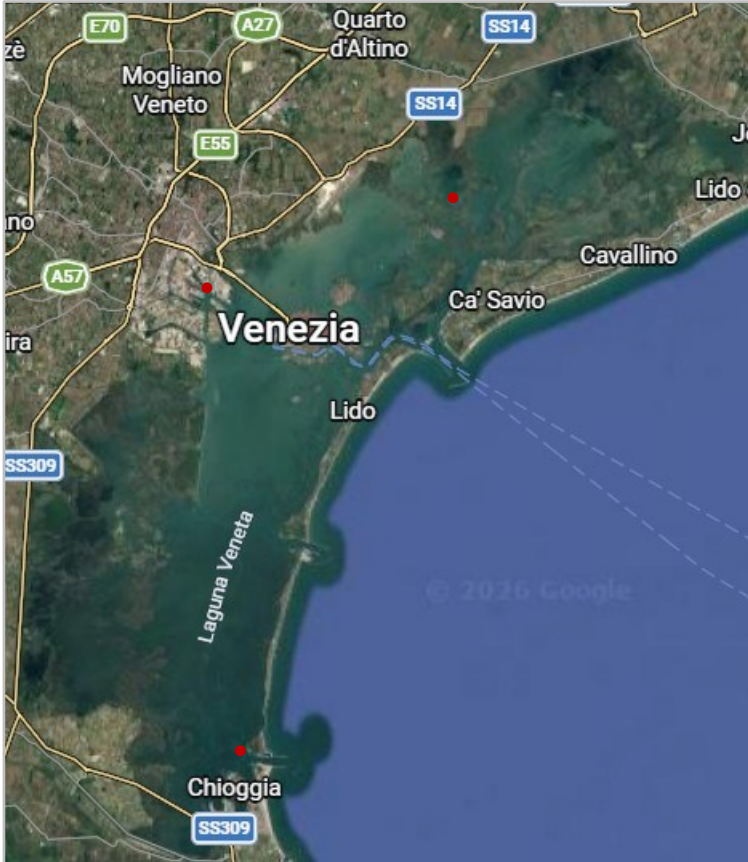
How to get metatranscriptomic data?

RNASeq is a good idea but, what about the rRNA?
(80-90% of RNA molecules are ribosomal RNAs)

Two different options:

- 1) Greatly increase the number of sequences
- 2) Remove the rRNA molecules somehow but,
how to easily remove the rRNA from samples?

A first metatranscriptomic study in the Venice Lagoon



Metatranscriptomic samples analysed

- 3 sampling sites
 - 5 seasons
 - 2 matrices per site
- = 29 different samples

Metatranscriptomic samples analysed										Sample Accession
Sample_ID	Box position	Sample_type	Date	Sample_name	Replica number	[RNA] (ng/μl)	[DNA] (ng/μl)	RIN	RNA/DNA	
• 1	B01	Sediment	20190213	BCHIO_sediment_20190213	1	61.2	2.1		29.1	ERS12293297
2	B02	Water	20190213	MARGH_water_20190213	1	75.2	32.4		2.3	ERS12293316
3	B03	Sediment	20190213	MARGH_sediment_20190213	1	102	12.4		8.2	ERS12293302
4	B04	Water	20190213	PROSA_water_20190213	1	91.4	33		2.8	ERS12293325
5	B05	Sediment	20190213	PROSA_sediment_20190213	1	104	12.6		8.3	ERS12293307
6	B06	Water	20190515	BCHIO_water_20190515	1	35.8	8.12		4.4	ERS12293312
7	B07	Sediment	20190515	BCHIO_sediment_20190515	1	98.8	9.2		10.7	ERS12293298
8	B08	Water	20190515	MARGH_water_20190515	1	55.6	34.2		1.6	ERS12293317
9	B09	Sediment	20190515	MARGH_sediment_20190515	1	88.6	10.9		8.1	ERS12293303
10	C01	Water	20190515	PROSA_water_20190515	1	35.4	3.26		10.9	ERS12293326
11	C02	Sediment	20190515	PROSA_sediment_20190515	1	164	49.8		3.3	ERS12293308
12	C03	Water	20190822	BCHIO_water_20190822	1	55.2	33.6		1.6	ERS12293313
13	C04	Sediment	20190822	BCHIO_sediment_20190822	1	156	46.2		3.4	ERS12293299
14	C05	Water	20190822	MARGH_water_20190822	1	82.2	32.4		2.5	ERS12293318
15	C06	Sediment	20190822	MARGH_sediment_20190822	1	130	26.6		4.9	ERS12293304
16	C07	Water	20190822	PROSA_water_20190822	1	78.8	56		1.4	ERS12293327
17	C08	Sediment	20190822	PROSA_sediment_20190822	1	174	53		3.3	ERS12293309
18	C09	Water	20191104	BCHIO_water_20191104	1	47.8	7.42		6.4	ERS12293314
19	E01	Sediment	20191104	BCHIO_sediment_20191104	1	64	2.54	5.6	25.2	ERS12293300
20	E02	Water	20191104	MARGH_water_20191104	1	59.4	8.72		6.8	ERS12293323
21	E03	Sediment	20191104	MARGH_sediment_20191104	2	53.4	2.72	7.5	19.6	ERS12293305
22	E04	Water	20191104	PROSA_water_20191104	1	45.6	11.2		4.1	ERS12293328
23	E05	Sediment	20191104	PROSA_sediment_20191104	1	95.2	9.9		9.6	ERS12293310
24	E06	Water	20200217	BCHIO_water_20200217	1	71.2	16		4.5	ERS12293315
25	E07	Sediment	20200217	BCHIO_sediment_20200217	1	106	15.6	6.5	6.8	ERS12293301
26	E08	Water	20200217	MARGH_water_20200217	1	70.8	18.6		3.8	ERS12293324
27	E09	Sediment	20200217	MARGH_sediment_20200217	1	90.2	9.38	7.7	9.6	ERS12293306
28	F01	Water	20200217	PROSA_water_20200217	1	64.4	37.2		1.7	ERS12293329
29	F02	Sediment	20200217	PROSA_sediment_20200217	2	102	13.6	7.2	7.5	ERS12293311

How was the RNA processed and how were the RNASeq data analysed?

- AllPrep DNA/RNA/miRNA Universal kit —————> water samples
- RNeasy PowerSoil Total RNA kit plus —————> sediment samples
PowerSoil DNA elution kit
- rRNA removal with a mixture of QIAseq FastSelect kits
- Illumina stranded RNAseq library preparation and sequencing
- Data uploaded in European Nucleotide Archive (ENA)
- Data made available to EBI-MGnify for assembly and annotation

Stuck

 General Error

Something went wrong while loading this page. Please refresh the page and if the error persists, let us know at metagenomics-help@ebi.ac.uk

Final considerations

(digging into my memory & today work)

- Many contigs, most of them short, few retained for length
- Higher number of retained contigs in water samples
- PKS transcripts are there
- In Marghera site, the metabolic pathways related to “Aromatics degradation” are much more represented.
- I need some time to truly analyse the results!!!